Date – November 26, 2025

Task 02 – Descriptive Analytics on Insurance Dataset

Objective:

To perform descriptive analytics focusing on variance, standard deviation, and all related statistical concepts, using a real-life inspired insurance dataset.

Problem Background:

An insurance analytics team in the USA wants to understand the risk pattern of customers. Variation in customer age, premiums paid, and claim amounts helps insurance companies decide:

- How to set premium prices
- How to predict future claims
- How to segment customers
- How to identify high-risk customers

You are given a dummy dataset of 300 insurance customers. Your job is to compute detailed measures of variability.

Detailed Dataset Description – Insurance Customer Dataset

This dataset contains **300 individual insurance customer records** from a fictional U.S.-based insurance company. It includes demographic data, policy information, and claim-related financial details. The dataset is suitable for descriptive analytics, variance analysis, risk modeling, and premium optimization studies.

1. Customer_ID

- **Type:** Numeric (Integer)
- **Range:** 1 to 300
- **Description:**
  A unique identifier assigned to each customer.
  It does not carry analytical meaning but is essential for indexing, referencing, or merging data.

2. Age

- **Type:** Numeric (Integer)
- **Range (based on generated data):** 18 to 74 years
- **Purpose in analytics:**
  - Helps insurers determine risk level (e.g., **older customers may have higher health/life claims**).
  - Used in customer segmentation and pricing models.
- **Expected distribution:**
  Uniform-like distribution across adult age groups.

3. Policy_Type

- **Type:** Categorical
- **Categories:**
  - **Health**
  - **Auto**
  - **Home**
  - **Life**
- **Purpose in analytics:**
  - Shows what type of insurance each customer holds.
  - Important for grouping customers and comparing claim patterns between policy categories.
- **Real-life use:**
  Auto insurance tends to have more frequent claims; Life/Health may have high-value claims.

4. Annual_Premium

- **Type:** Numeric (Integer)
- **Range:** $300 to $5000
- **Description:**
  Annual amount paid by the customer for their insurance policy.
- **Significance in analytics:**
  - Indicates **policy level** (basic, standard, premium tier).
  - Variability helps identify market segmentation.
  - Premium variance can highlight:
    - Income differences
    - Type of policy
    - Customer risk level

5. Claims_Last_Year

- **Type:** Numeric (Integer)
- **Range:** 0 to 4
- **Explanation:**
  Number of insurance claims filed by the customer in the previous year.
- **Meaning:**
  - Higher values may suggest high-risk customers.
  - Useful for predicting future claim probability.
- **Expected pattern:**
  Most customers have **0 or 1 claim**, as real-world claim frequency is usually low.

6. Claim_Amount

- **Type:** Numeric (Integer)
- **Range:** $0 to $20,000
- **Description:**
  Total dollar value of claims filed by the customer last year.
- **Interpretation:**
  - $0 indicates no claims.
  - Higher values represent severe or multiple incidents.
- **Analytical importance:**
  - Core metric for insurance risk assessment.
  - Used for variance analysis, outlier detection, risk scoring.

Your Task Instructions(Google Colab)

1. Upload the dataset

2. Perform Descriptive Analytics (A–Z of Variance & Standard Deviation)

A. Basic Statistical Summary

B. Compute Variance & Standard Deviation

C. Explain These Concepts Briefly

D. Plot Visualizations

E. Interpret the Variance

F. Final Report